

Ethan MacLeod

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EDUCATION

The University of Waikato

Bachelor of Engineering with Honours (BE HONS), Majoring in Software Engineering

Hamilton, New Zealand

Completed 2024

Waikato Institute of Technology

WINTEC Manufacturing and Technology Vocational Pathway

Hamilton, New Zealand

Completed 2019

TECHNICAL SKILLS

Languages: JavaScript, TypeScript, Python, Java, C#, SQL, HTML5, CSS3

Frontend: React, React Native, Svelte, SvelteKit, Redux, Responsive Design

Backend: Node.js, Django, Express.js, REST APIs, Database Design, PostgreSQL, API Development

Developer Tools: AWS, Docker, Terraform, Git, CI/CD, Linux, Figma, LaTeX, Agile, Microservices

EXPERIENCE

Contract Software Engineer

2025 – Present

Bearing

Remote

- Built **7 full-stack applications** using **SvelteKit**, **TypeScript**, and **PostgreSQL** for industry clients
- Collaborated with **4 cross-functional teams** including designers and data engineers on scalable development
- Engineered data integration solutions processing **100K+ records** daily with automated transformation pipelines
- Implemented authentication and **10+ external API** integrations supporting user security

Software Engineer

2023 – Jan 2025

Ahuora – Centre for Smart Energy Systems

Hamilton, New Zealand

- Collaborated with **3 universities** and **15 engineers** internationally on application development
- Delivered simulation platform serving **20+ industry experts** in process heat optimization research
- Architected scalable PostgreSQL database with Django ORM supporting distributed research environments
- Integrated IDAES **nonlinear programming** solvers to process industrial datasets
- Led frontend development using **React**, **TypeScript**, and **RTK Query**, reducing data loading times by **60%**

PROJECTS

Micro-Grid Simulator | *Django, React, TypeScript, PostgreSQL, Docker* | [GitHub](#)

2024

- Built interactive web platform for energy system modeling with **real-time visualization** of micro-grid energy flows and resource allocation
- Developed full-stack application using **Django REST Framework** backend and **React/TypeScript** frontend with **50+ components**
- Implemented customizable simulation scenarios enabling users to modify energy inputs, consumption patterns, and environmental factors
- Containerized application with **Docker** supporting **PostgreSQL** database and caching for energy calculations

Pinch Analysis | *Python, Visual Basic, Figma, NLP* | [GitHub](#)

2024

- **Recipient of the 2024 Staff Choice Best Engineering R&D Project**
- Migrated legacy Visual Basic code to Python for enhanced performance and maintainability
- Integrated complex process engineering modules into a functional tool
- Conducted user testing to inform frontend design, improving usability and aesthetics
- Achieved a 70% performance improvement in calculation time over existing Pinch Analysis tools (OpenPinch)

BetterReads Cloud Platform | *AWS, Terraform, Docker, CI/CD, Node.js* | [GitHub](#)

2023

- Architected AWS infrastructure using Terraform supporting **4 availability zones** with **99.9%** uptime
- Implemented **IAM** security policies and **VPC** networking protecting **5 microservices**
- Configured multi-tier architecture with **6 subnets** (public/private) for optimal resource isolation and performance
- Built automated CI/CD pipeline reducing manual deployment time from **45 minutes to 8 minutes** with integrated testing and rollback capabilities